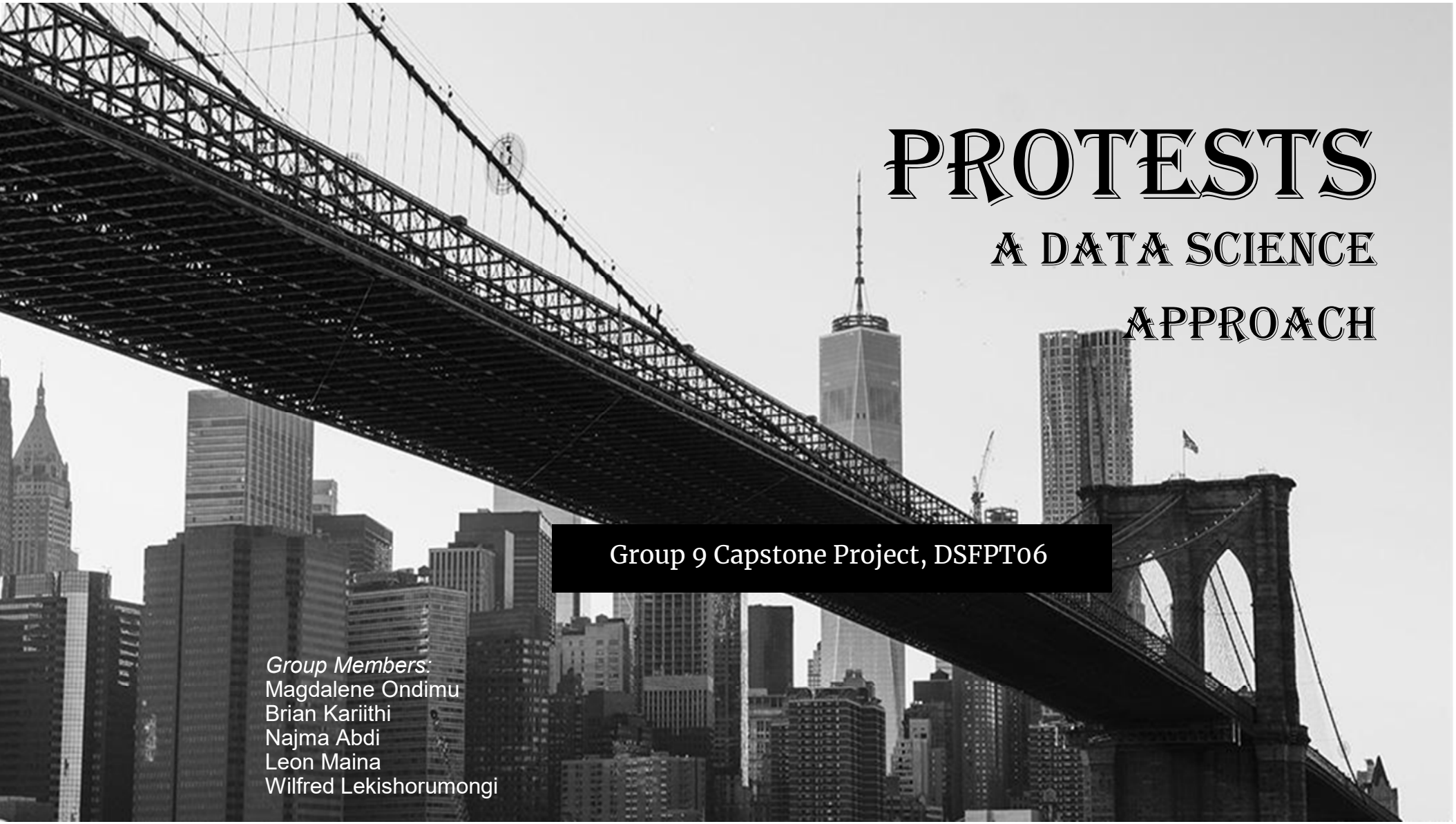

PROTESTS

A DATA SCIENCE APPROACH

Group 9 Capstone Project, DSFPT06

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PREDICTING STATE RESPONSES TO PROTESTS

A Machine Learning Approach Using Topic Modelling, Sentiment Analysis, and Classification Using Key Protest Attributes

OVERVIEW

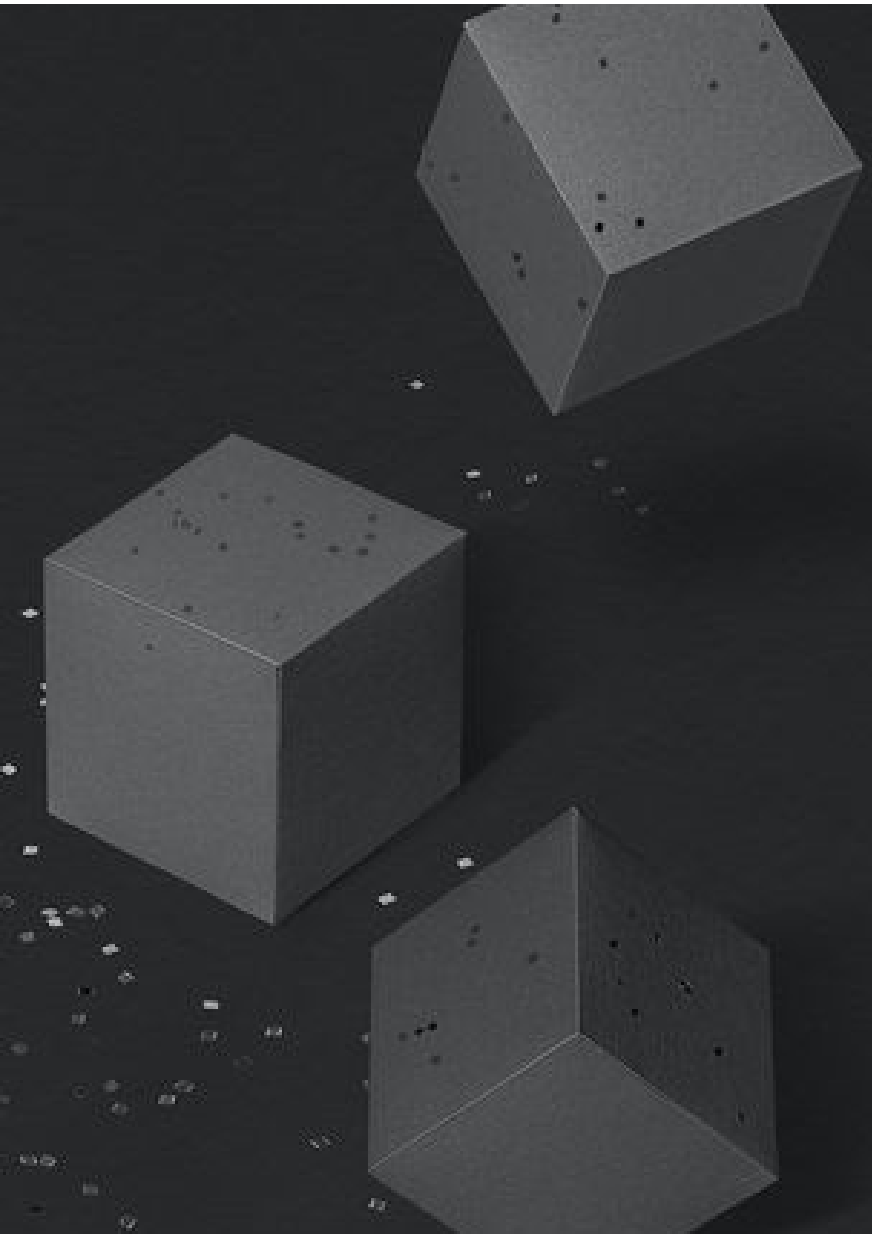
The project aims to analyze global protest events to understand their underlying causes, geographical and temporal trends, patterns, and state responses.

This comprehensive analysis is crucial for informing policy and decision-making to manage socio-political unrest effectively.

PROBLEM STATEMENT

Understanding and predicting state responses to protests is crucial for improving state-citizen relations and managing social unrest.

This project aims to provide valuable insights into public sentiment during protests and help policymakers and organizations make informed decisions. By leveraging machine learning, we can uncover patterns and trends in protest-related data, enabling more proactive and effective responses.



MAIN OBJECTIVES

- Develop and evaluate machine learning models to predict state responses to protests based on the identified sentiments, topics and other protest attributes such as region, protest duration and protest identity.
- Conduct sentiment analysis on protest notes and tweets to classify sentiments as positive, negative, or neutral.
- Identify and summarize key topics from protest-related communications.

SPECIFIC OBJECTIVES

- Enhance data preprocessing and feature engineering techniques to improve model performance.

- Perform dimensionality reduction to handle high-dimensional data and reduce overfitting.

- Fine-tune hyperparameters and explore ensemble methods to optimize model accuracy.

DATA COLLECTION

Description:

- Collected data from the Havard mass mobilization dataset https://dataverse.harvard.edu/file.xhtml?persistentId=doi:10.7910/DVN/HTTWYL/TJJZNG

- Collected and Analyzed Tweets from Recent Protest Activities (July 13th to July 23rd 2024)

Data Cleaning: Handled missing values, removed duplicate values, filled null values, converted textual data to numerical data among others.

DATA UNDERSTANDING

Dataset Overview:

- **Source**: The dataset is sourced from the Mass Mobilization Project at Harvard, which records global instances of protests and state responses.

- **Time Period**: Covers a wide range of years from 1990-2020, providing a longitudinal view of protest dynamics.

- **Event Details**: Includes event dates,size of the protest,countries and regions of protests.

Tweets Key Features :

- **Date**: The date when the tweet was posted.

- **Sentiment:** The sentiment expressed in the tweet, which could be positive, negative, or neutral.

CURRENT PROTESTS FROM AROUND THE WORLD

A protestor in Toronto, Canada carrying a placard highlighting a pursuit for freedom of the Palestinians and a stop to genocidal killings.

Anti-government protesters display Bangladesh's national flag as they storm Ex Prime Minister Hasina's palace.

CURRENT PROTESTS FROM AROUND THE WORLD

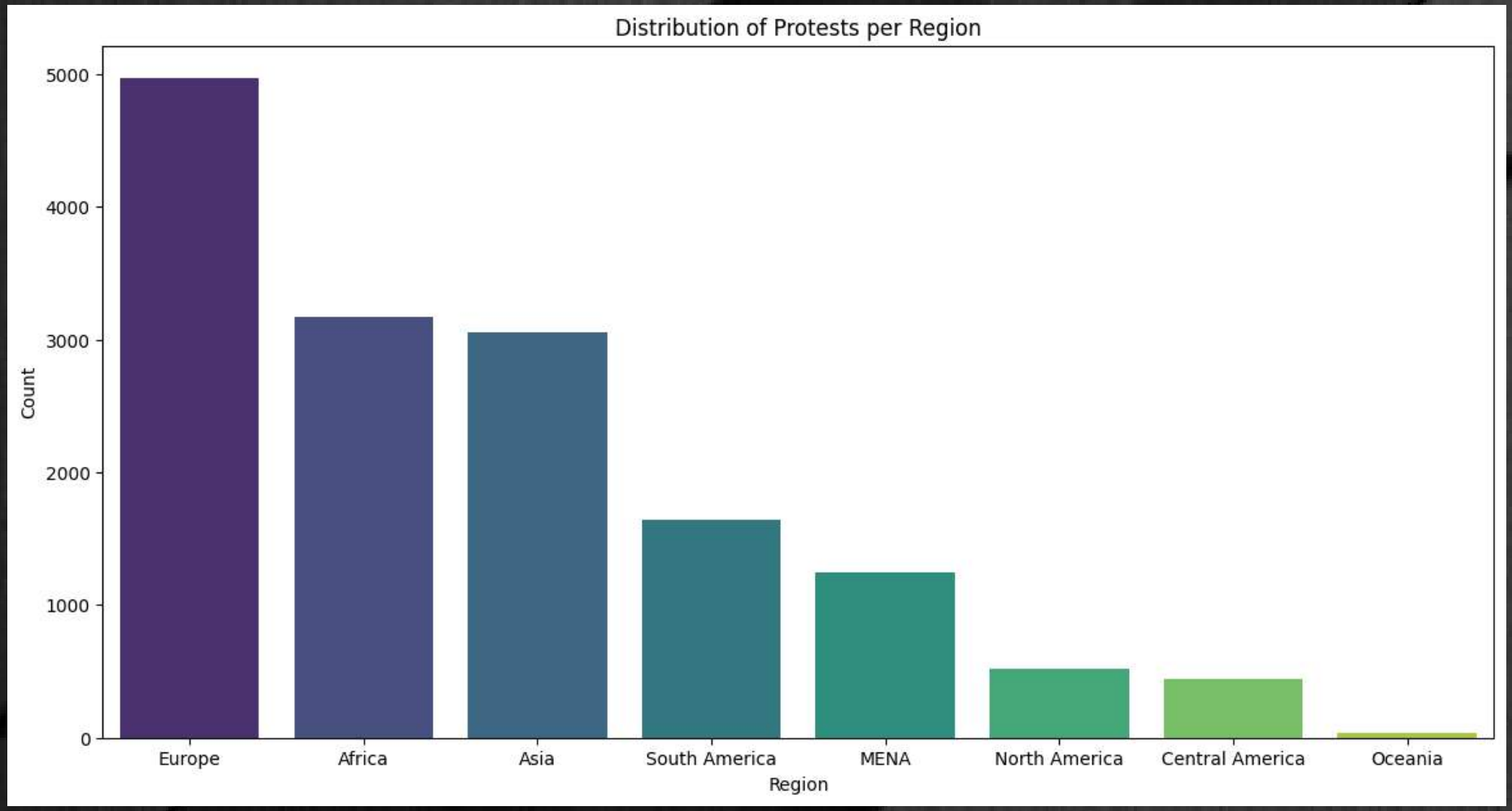
Protesters hold a Kenyan flag outside the Kenyan Parliament after storming the building during a nationwide strike to protest against tax hikes and the Finance Bill 2024 in downtown Nairobi, on June 25, 2024.

Protesters make signs with their arms in front of Kenya police officers during a demonstration against tax hikes as Members of the Parliament debate the Finance Bill 2024 in downtown Nairobi, on June 18, 2024.

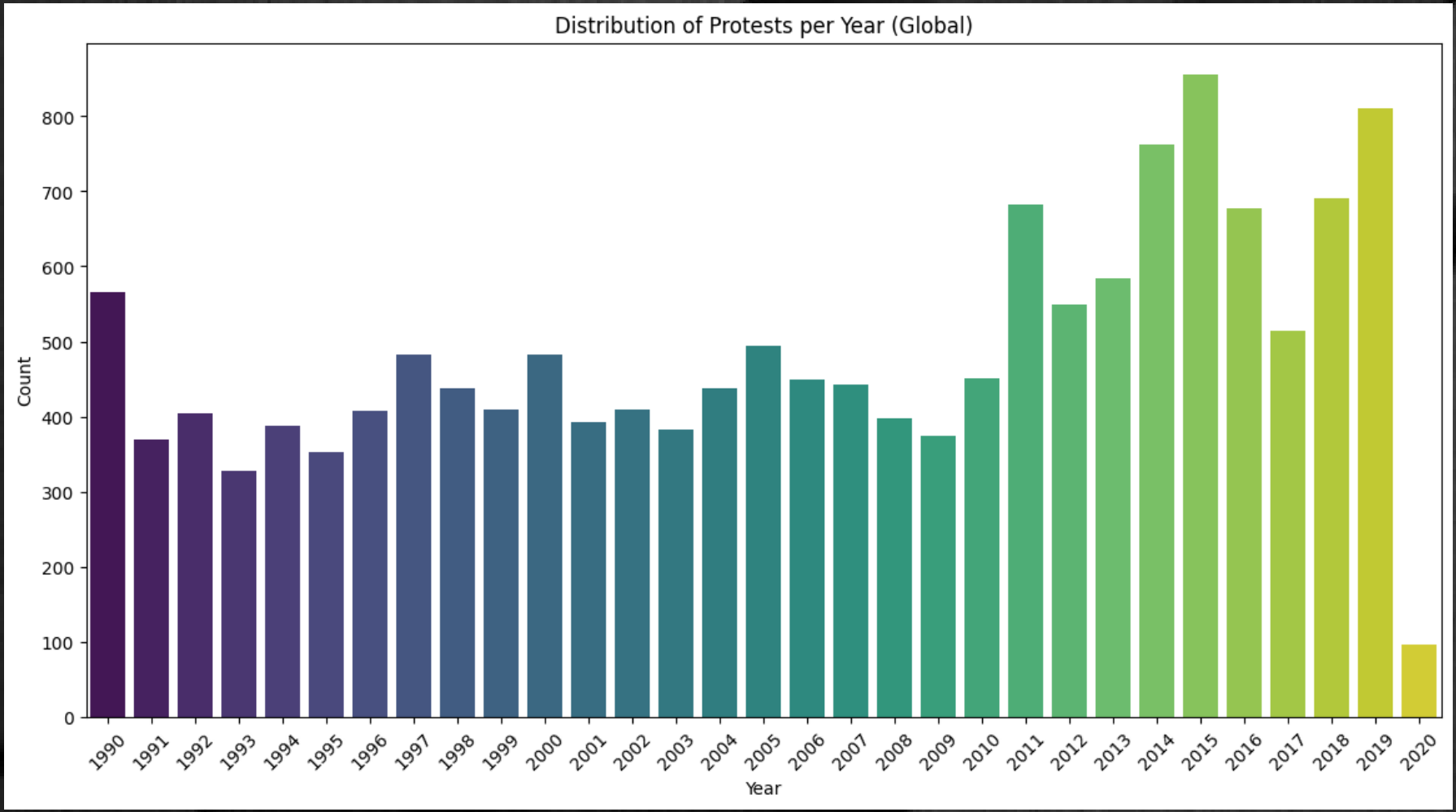
EXPLORATORY DATA ANALYSIS (EDA)

Data Insights: Provided insights into the distribution and nature of the data.

Visualizations: Created histograms, box plots, bar graphs, line plots, scatter plots and wordclouds for better understanding of the data.

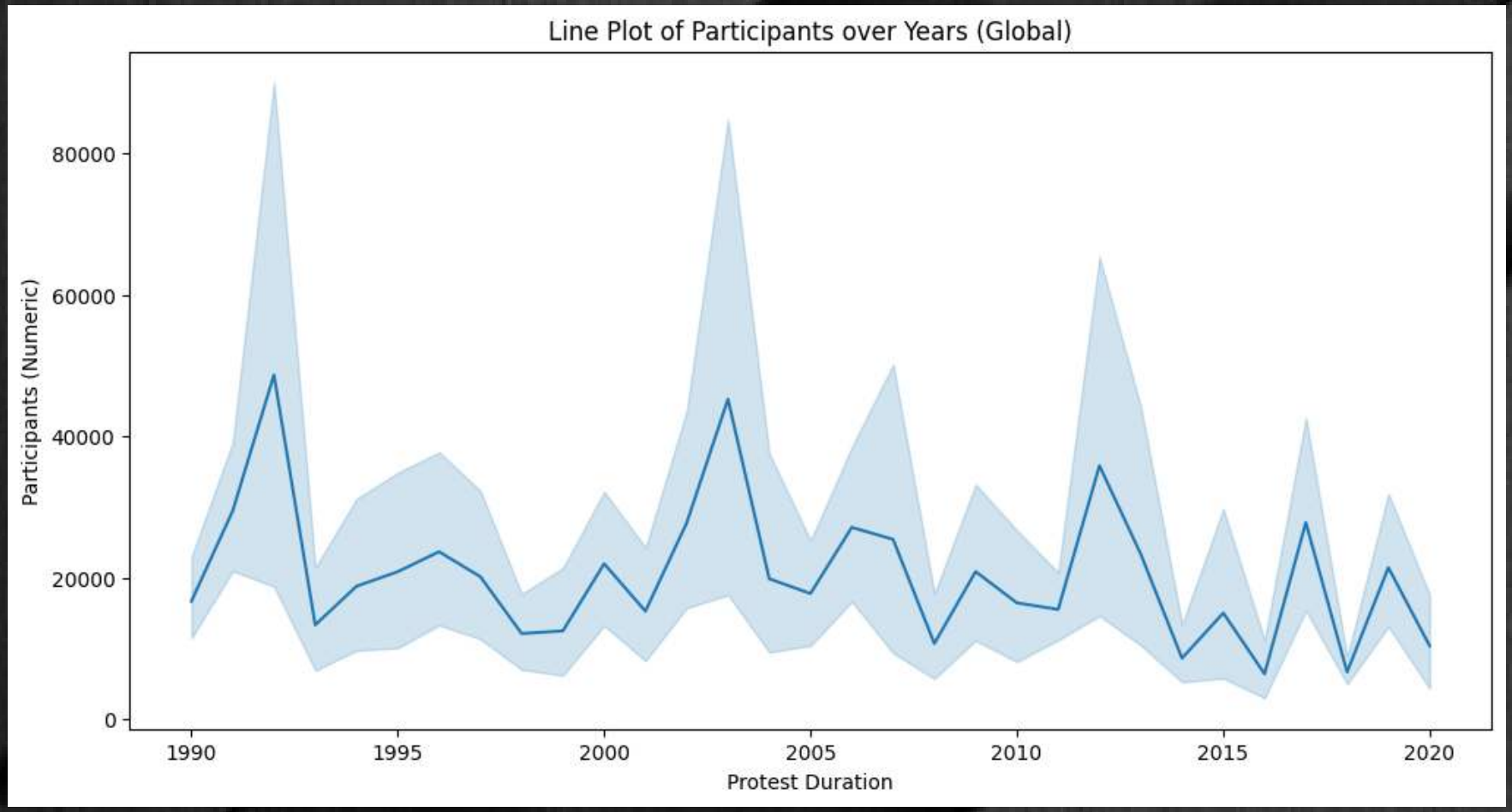


- Europe had the most recorded protests ,Africa being the second and Oceania recording the least.

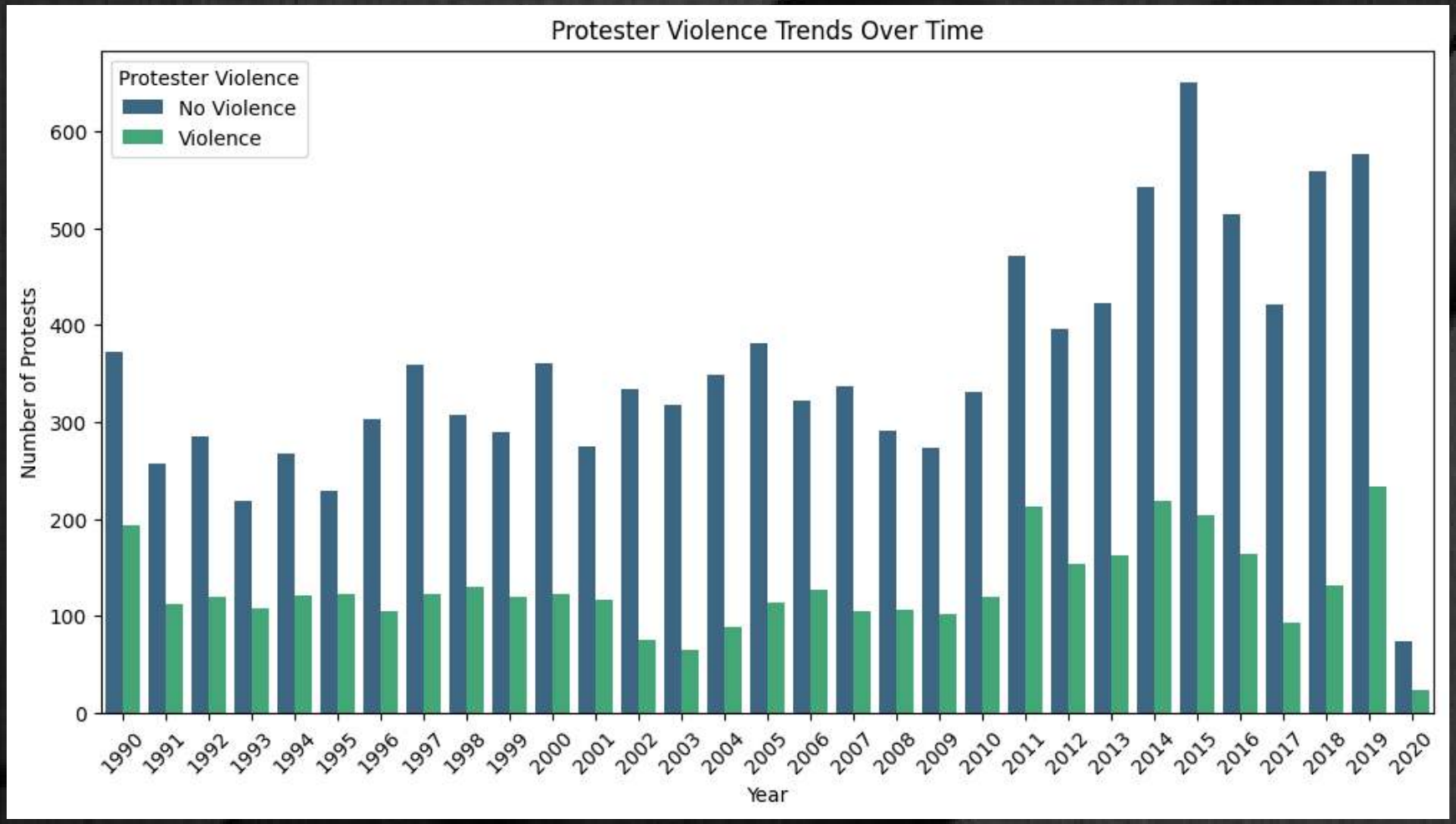


The distribution of protests took an upward turn from 2010's up until 2020 likely influenced by the global COVID pandemic

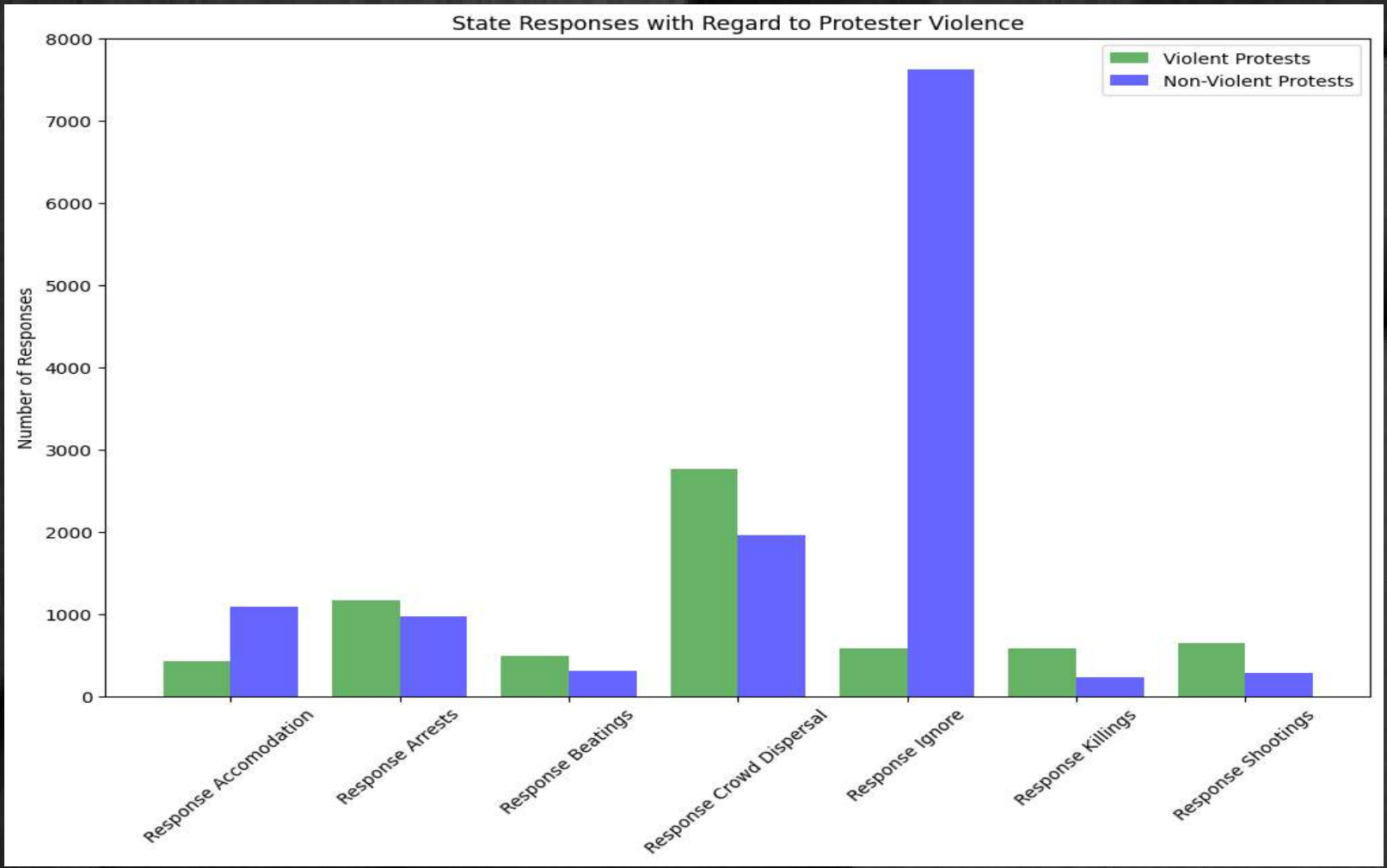
Kenya had a record high number of protests in 2015.This was also observed in other nations.



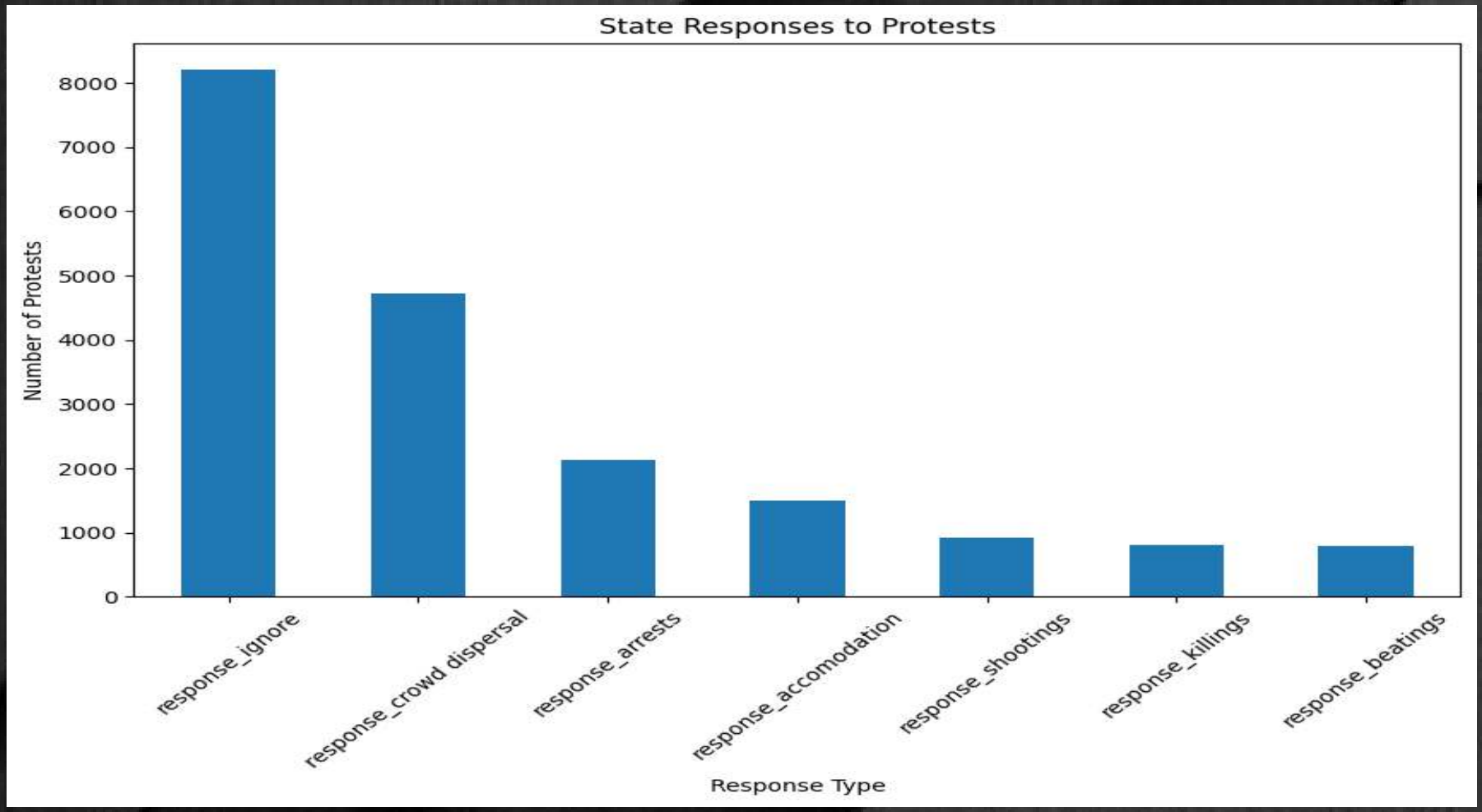
- There are noticeable spikes around the early 1990s, late 1990s, and mid-2000s, suggesting significant global protest events during these periods.
- After the mid-2000s, the number of participants in protests seems to decrease overall, with smaller spikes occurring in the 2010s.



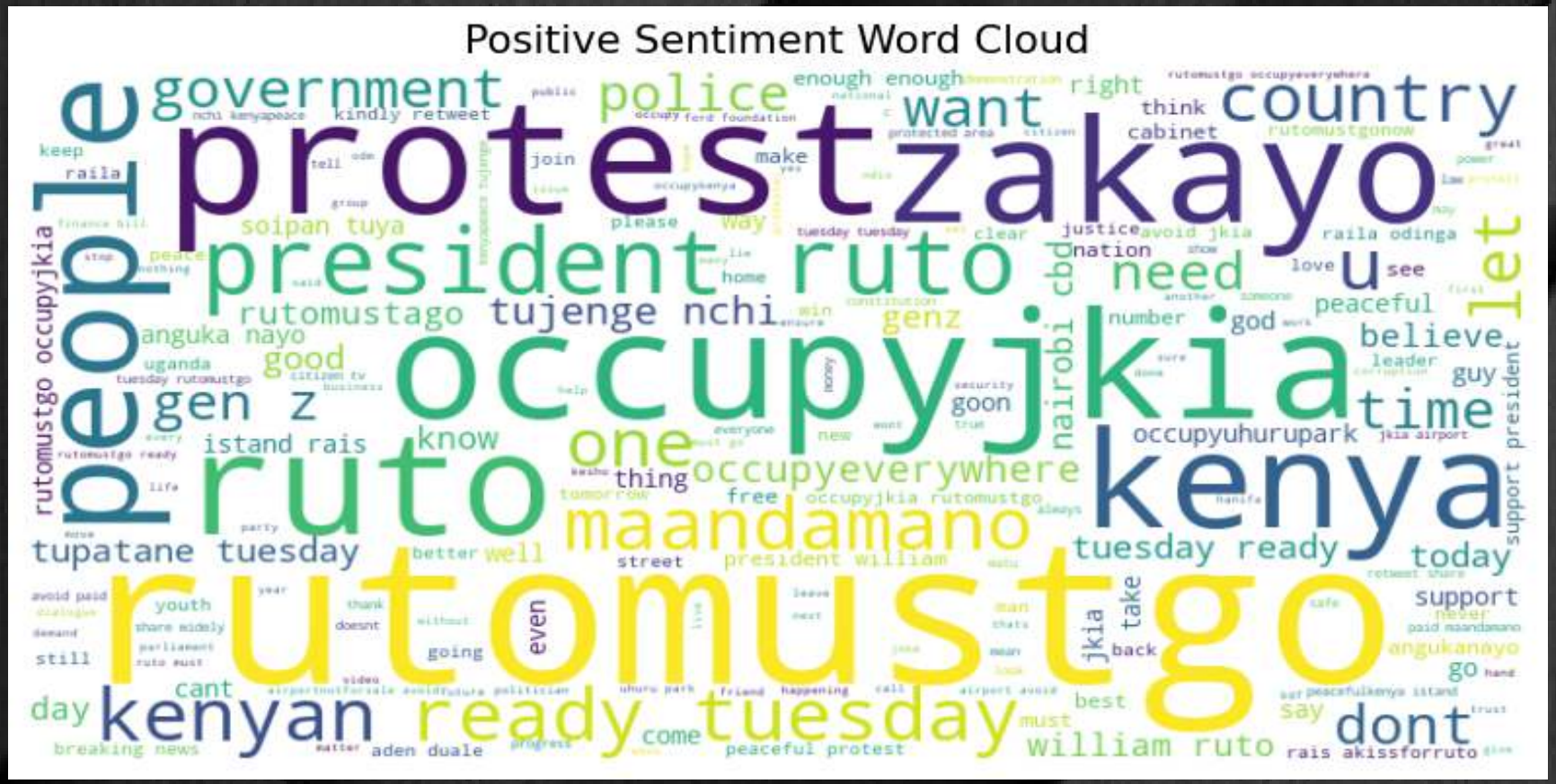
In 2015, the highest number of protests without violence was recorded, surpassing 600 incidents. Conversely, in 2019, the highest number of violent protests was observed, exceeding 200 incidents.

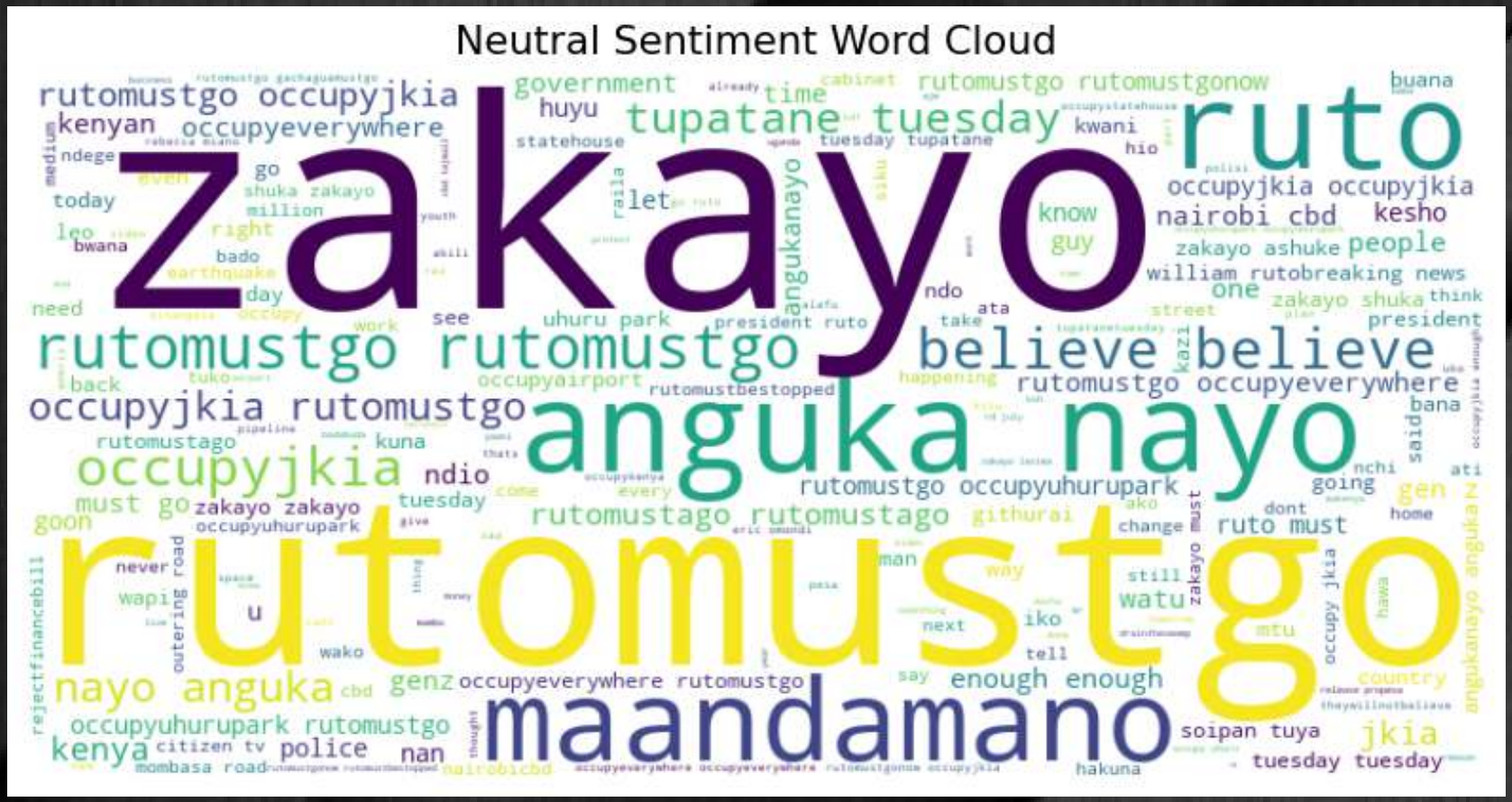


- Violent Protests: Frequent, forceful; "Severe Intervention" common
- Non-Violent Protests: Less frequent, less forceful; "Moderate Intervention" most common
- Non-Intervention: Rare overall



- Forceful Responses: Most common: Arrests; then Shootings, Killings, Beatings, Crowd Dispersal
- Non-Confrontational Responses: Most common: Ignoring; then Accommodation


Positive Sentiment Word Cloud
protest
zakayo
rutomustgo
occupyjkia
president
ruto
people
government
police
want
country
kenya
maandamano
tuesday
ready
kenyan
occupyeverywhere
tujenge nchi
rutomustago
need
time
let
u
believe
peaceful
gen z
good
one
know
istand rais
tupatane tuesday
tuesday ready
today
support
dont
william ruto
day
cant
even
nairobi
cbd
jkia
take
back
genz
goon
occupyuhurupark
rutomustgo occupyjkia
enough enough
right
think
cabinet
rutomustgonow
raila odinga
justice
avoid jkia
nation
love
see
number
god
guy
president
soipan tuya
anguka nayo
uganda
tuesday rutomustgo
thing
free
occupyjkia rutomustgo
street
president william
better
well
youth
share widely
doesnt
going
still
aden duale
peaceful protest
come
must
best
say
rais akissforruto
angukanayo
go
peacefulkenya istand
paid maandamano
never
breaking news
kindly retweet
nchi kenyapeace
raila
keep
join
make
please
way
tell
occupykenya
ford foundation
protected area
citizen
power
great
tuesday tuesday
clear
home
continuation
another
someone
leader
security
done
sure
help
everyone
true
new
must go
life
party
year
thank
without
live
leave
next
man
thats
mean
look
friend
happening
call
airport avoid
uhuru park
politician
airportnotforsale
avoid future
video
trust
rutomustgo occupyeverywhere
rutomustgo ready
tuesday rutomustgo
citizen tv
business
money
jkia airport
support president
rutomustgo occupyjkia
nchi kenyapeace
finance bill
stop
peace
nothing
said
group
tuesday
tuesday
tuesday


Neutral Sentiment Word Cloud
zakayo
ruto
rutomustgo
maandamano
anguka
nayo
rutomustgo
rutomustgo
believe
believe
occupyjkia
occupyjkia
tuesday
tupatane
government
rutomustgo occupyjkia
kenyan
occupyeeverywhere
nairobi cbd
kesho
people
occupyjkia occupyjkia
rutomustgo occupyeverywhere
rutomustgo occupyuhurupark
rutomustago
rutomustago
ruto must
nayo anguka
occupyuhurupark rutomustgo
kenya
police
nan
genz
occupyeverywhere rutomustgo
enough enough
soipan tuya
jkia
tuesday tuesday
hakuna
watu
iko
mtu
country
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president ruto
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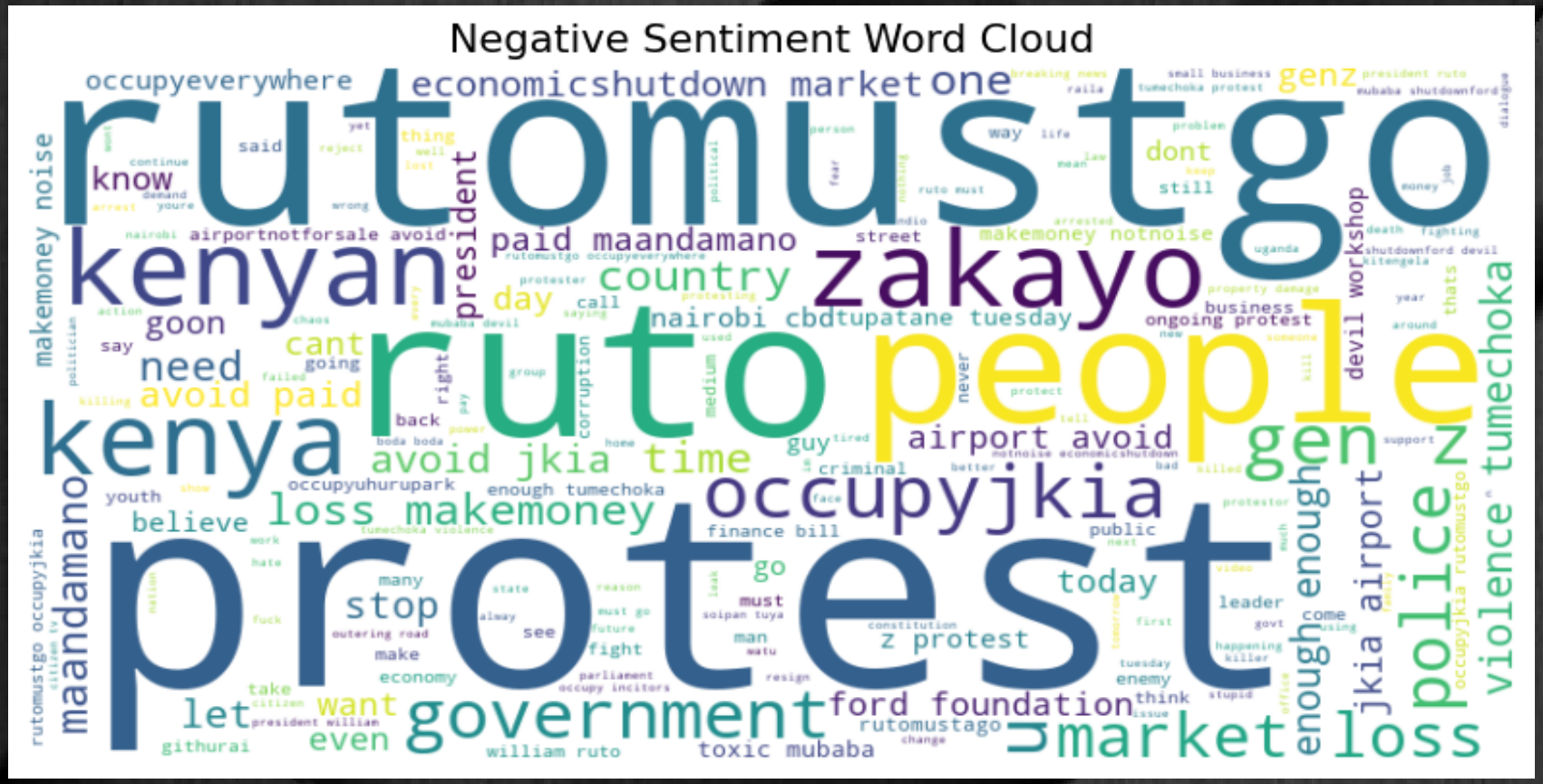
Negative Sentiment Word Cloud
rutomustgo
protest
kenyan
ruto
people
kenya
zakayo
occupyjkia
government
market
loss
police
violence
tumechoka
maandamano
makemoney
noise
occupyeverywhere
economicshutdown
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said
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way
airportnotforsale
avoid
paid
maandamano
president
country
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nairobi
cbdtupatane
tuesday
makemoney
notnoise
goon
need
cant
avoid
paid
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youth
criminal
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corruption
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protest
public
leader
enemy
think
stupid
many
state
must
man
make
economy
fight
see
outering road
take
parliament
occupy inciters
president willian
william ruto
githurai
constitution
tuesday
happening
killer
govt
come
using
first
future
always
must go
soipan tuya
watu
resign
change
issue
office
tomorrow
video
much
next
face
protester
support
bad
killed
better
hate
fuck
nation
work
show
tumechoka violence
citizen tv
rutomustgo occupyjkia
occupyjkia rutomustgo
dialogue
money
job
death
fighting
shutdownford devil
kitengela
uganda
property damage
protesting
protester
protect
tell
someone
kill
killing
action
chaos
mubaba devil
saying
used
boda boda
power
hoew
group
failed
going
say
politician
every
pay
nairobi
youre
demand
arrest
wrong
continue
reject
well
lost
thing
yet
person
fear
nothing
ruto must
political
law
life
problem
keep
mean
arrested
indio
street
small business
tumechoka protest
raila
breaking news
president ruto
mubaba shutdownford
dialogue
year
thats
around
someone
new
business
ongoing protest
kill
bad
better
killed
protester
much
next
video
leader
govt
happening
killer
tuesday
enemy
think
stupid
office
issue
change
resign
tomorrow
constitution
first
come
using
future
always
must go
soipan tuya
watu
man
make
fight
see
outering road
take
parliament
occupy inciters
president willian
william ruto
rutomustago
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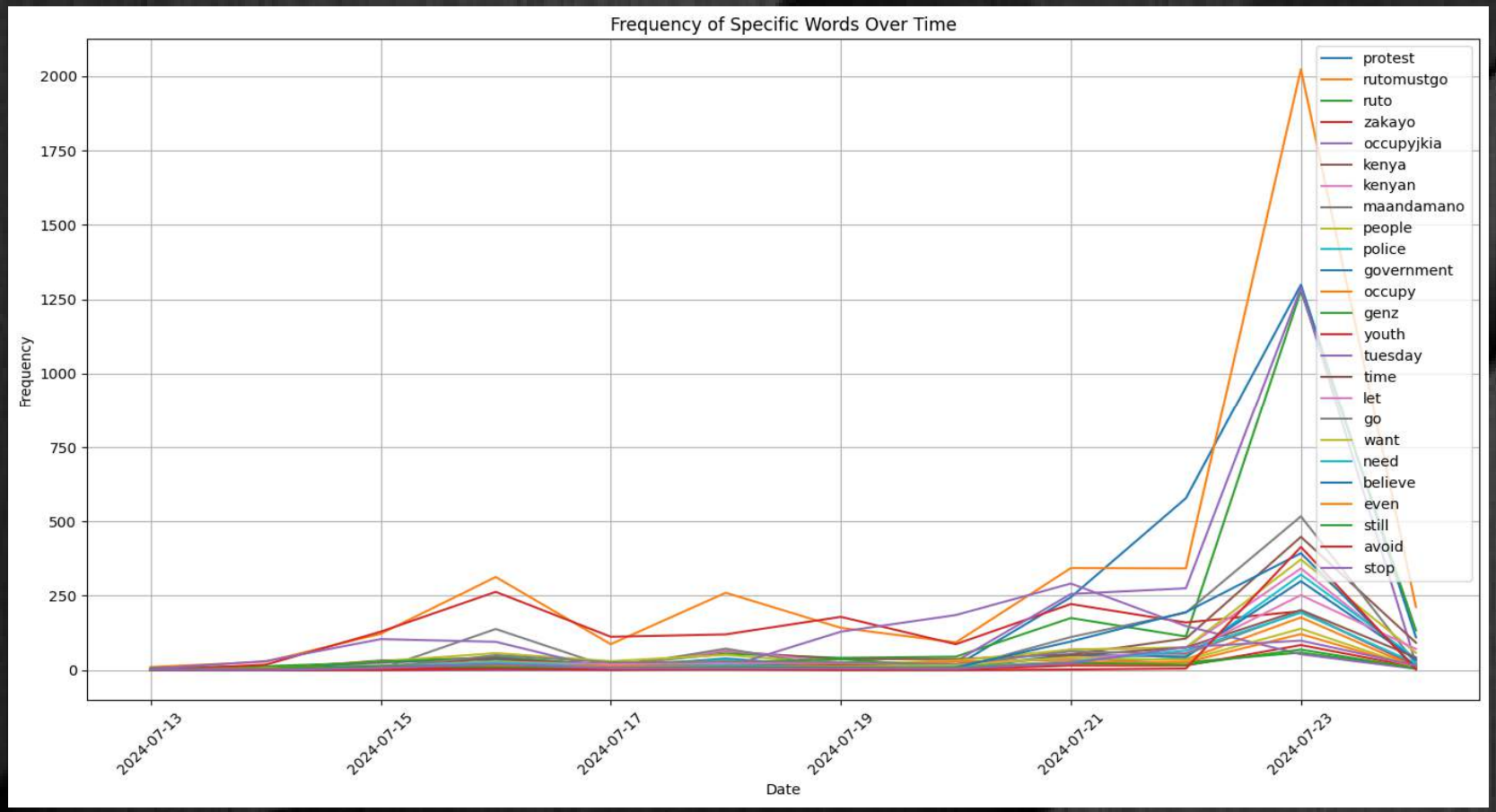
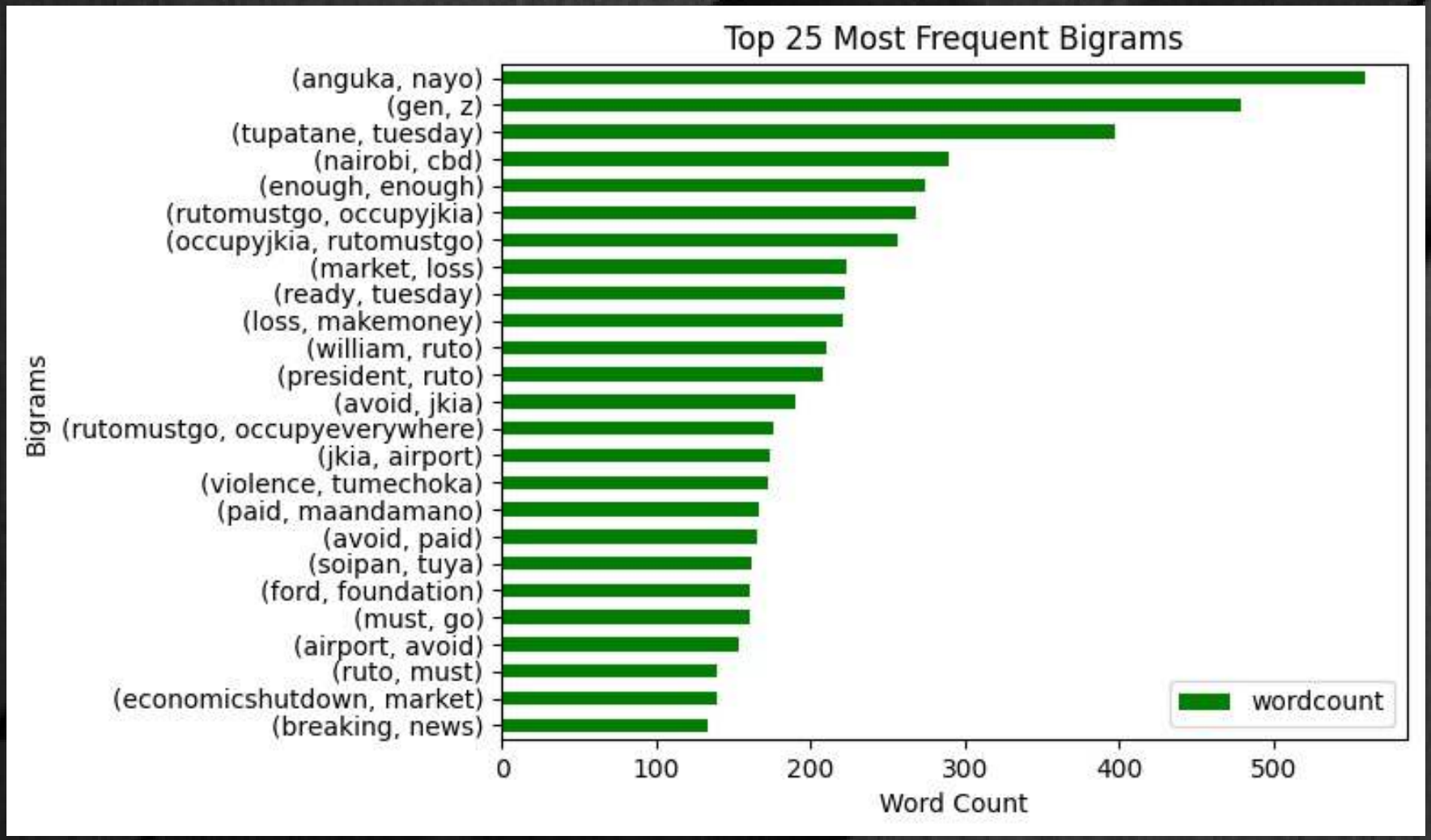


- Trends (July 13-23, 2024): Frequent terms include "protest," "rutomustgo," "ruto," "occupyjkia.“
- Insights: Fluctuations indicate event-driven interest.
- Emerging: Rising mentions of "genz" and "youth."

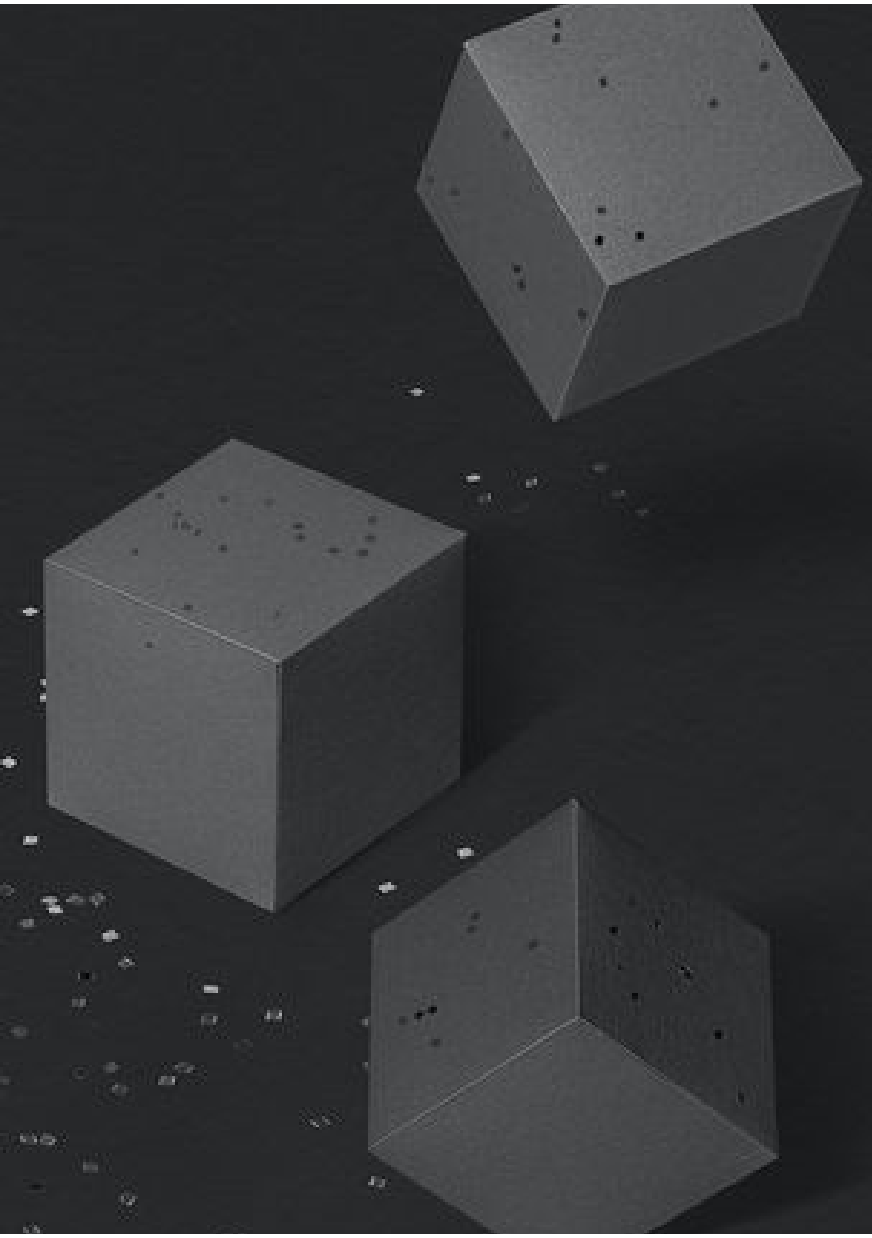


Bigram: Pair of consecutive words used together in text analysis

DATA PREPARATION

Steps:

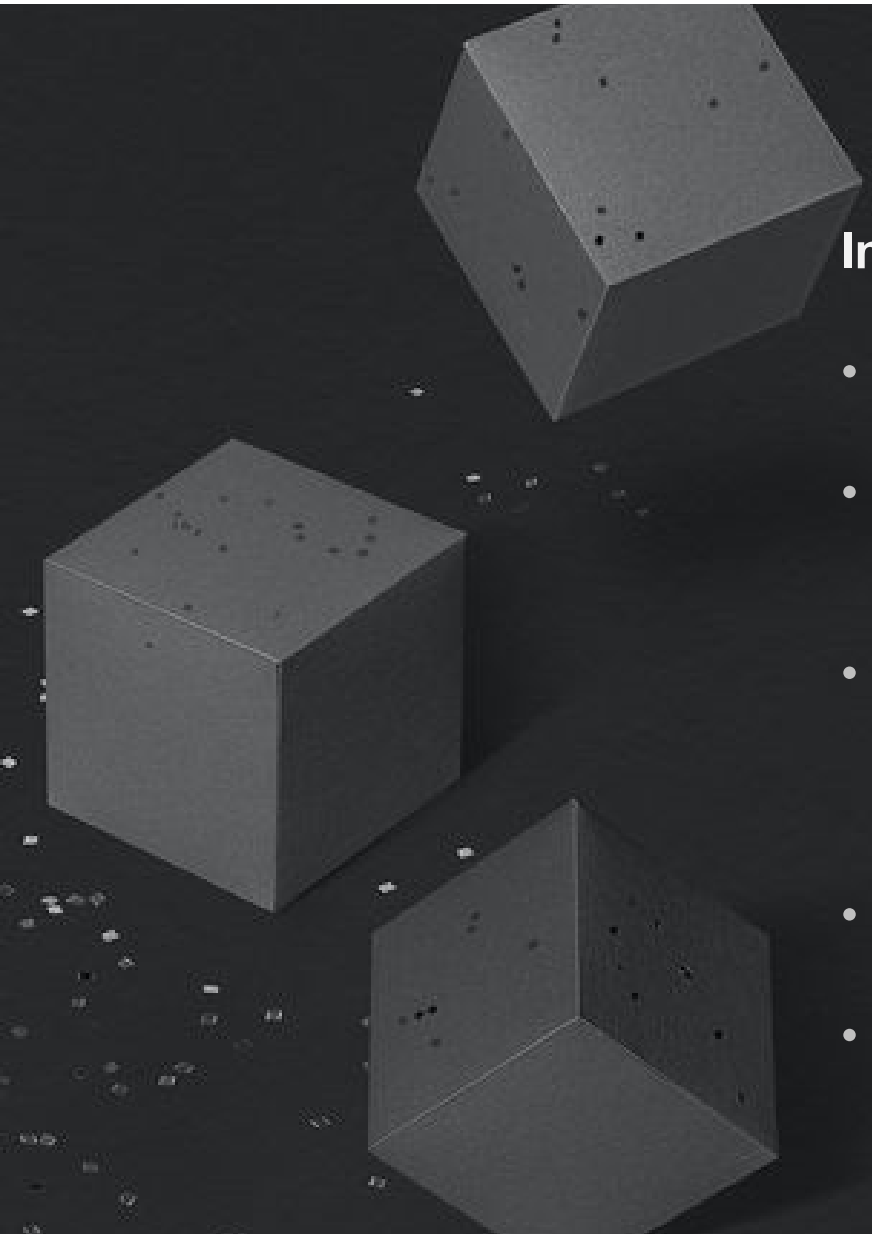
- Cleaned data by removing extreme values in protest duration and participant numbers column
- Region analysis: Combined Americas, excluded Canada and Oceania.
- Dropped country column.
- Grouped responses into three classes: Passive/Concessive, Control Measures, and Forceful Repression to enhance analysis and modeling.



BASELINE MODEL

Logistic Regression:

- Training accuracy: 52.9%.
- Test accuracy: 52.6%.
- Performance issues: The model is not accurately identifying all instances of Control Measures, leading to some being misclassified.



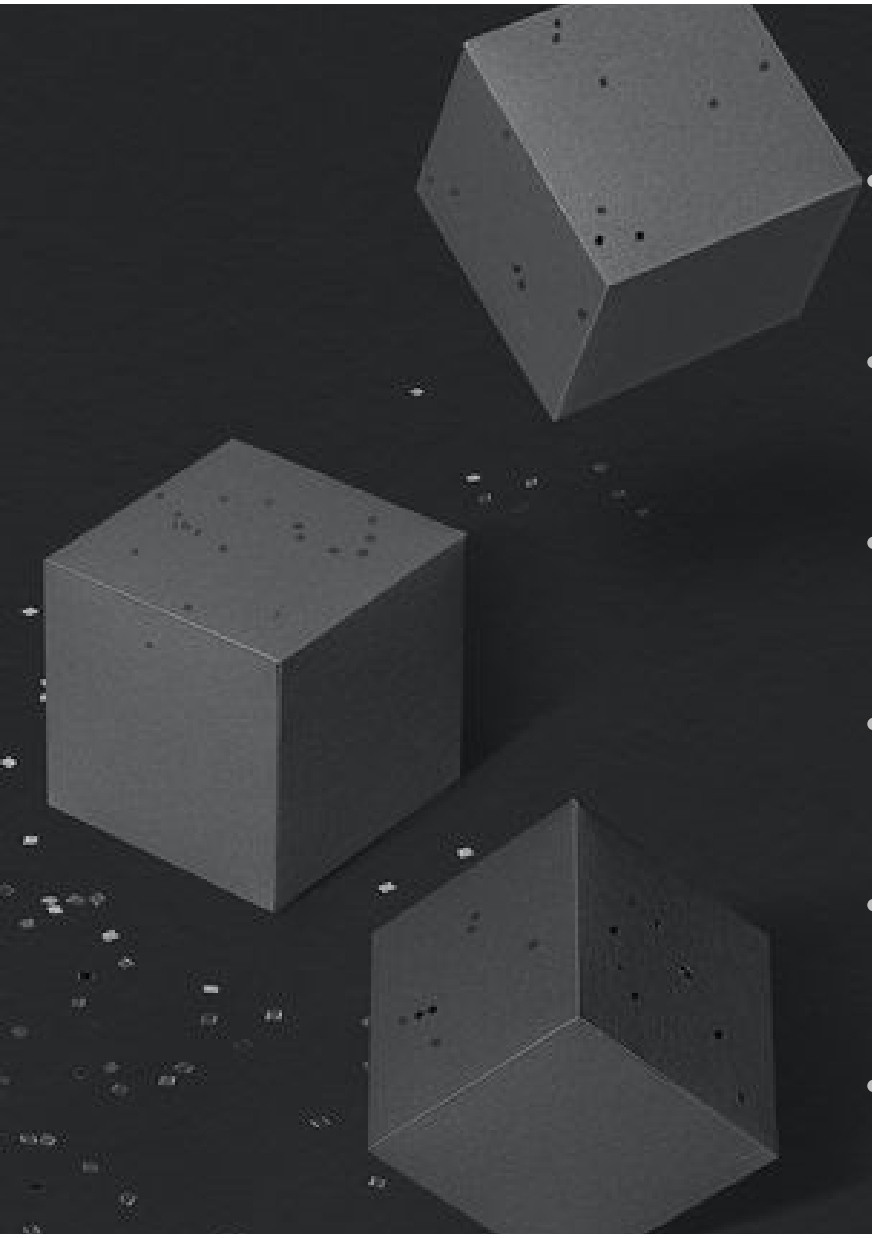
MAIN MODELS

Improvement Steps:

- Hyperparameter tuning.
- Explored XGBoost, GradientBoosting, Random Forest classifier and SVC.
- Feature engineering including LDA Topic Modelling, Sentiment Feature Extraction and Named Entity Recognition.
- Address class imbalance (SMOTE)
- Performed Linear Discriminant Analysis as a dimensionality reduction technique.

MAIN MODELS PERFOMANCE AFTER TUNING VS BASELINE MODEL:

Models	Train Accuracy	Test Accuracy
Baseline Model	52.9%	52.6%
Random Forest	1.0	0.79
XGBOOST	0.83	0.81
SVC	0.82	0.81



Evaluation Metrics

- Precision:
  True Positives / (True Positives + False Positives)
- Recall (Sensitivity):
  True Positives / (True Positives + False Negatives)
- F1 Score:
  Harmonic mean of precision and recall
- Confusion Matrix:
  Table of TP, TN, FP, FN
- ROC-AUC Score:
  Area under ROC curve
- Precision-Recall AUC:
  Area under precision-recall curve

BEST PERFORMING MODEL

- **Model:** XGBOOST CLASSIFIER

- **Training Accuracy:** 83%
- **Test Accuracy:** 81%

- **Classification Report:**

  Precision, Recall, F1 Score:
  Control Measures: Precision 0.76, Recall 0.74, F1 Score 0.75
  Forceful Repression: Precision 0.83, Recall 0.85, F1 Score 0.84
  Passive or Concessive: Precision 0.85, Recall 0.86, F1 Score 0.85

ROC AND PR CURVES

ROC Curves:

- Show trade-off between true positive rate (TPR) and false positive rate (FPR).
- AUC values indicate model performance:
  - Control Measures: AUC = 0.90
  - Forceful Repression: AUC = 0.96
  - Passive or Concessive: AUC = 0.96

PR Curves:

- Display precision vs. recall.
- High AUC values show effective identification:
  - Control Measures: AUC = 0.83
  - Forceful Repression: AUC = 0.91
  - Passive or Concessive: AUC = 0.92

CONFUSION MATRIX:

True Positives (Diagonal Elements): Correctly predicted instances.

- **Control Measures:** 1335
- **Forceful Repression:** 1494
- **Passive or Concessive:** 1471

False Positives and False Negatives (Off-Diagonal Elements): Instances where the model misclassified the labels.

- Misclassified as **Control Measures:** 244 (from Forceful Repression), 231 (from Passive or Concessive)
- Misclassified as **Forceful Repression:** 239 (from Control Measures), 30 (from Passive or Concessive)
- Misclassified as **Passive or Concessive:** 182 (from Control Measures), 63 (from Forceful Repression)

SUMMARY:

- The XGBoost model demonstrates robust classification performance with high accuracy and an excellent ability to distinguish between classes.

- The high AUC values across ROC and PR curves indicate the model's strong discriminative power, and the confusion matrix reflects a high number of correct predictions with relatively few misclassifications.

- This analysis underscores the model's effectiveness in classifying instances across the given categories

RECOMMENDATIONS:

- Further Model Tuning: Further hyperparameter tuning and explore other algorithms like ensemble methods to potentially boost performance further. Implement advanced techniques such as ensemble learning (e.g., stacking models) to leverage the strengths of multiple algorithms.

- Regular Model Evaluation: Continuously evaluate the model with new data to ensure its performance remains robust over time. Regularly update the model with recent data to maintain its accuracy and relevance.

- Feedback Loop: Establish a feedback loop with stakeholders to incorporate their insights and concerns into model improvements and data collection efforts.

CONCLUSION:

- High Model Performance: The XGBoost model showed excellent accuracy and effectiveness in distinguishing state responses, particularly excelling in detecting Forceful Repression and Passive or Concessive responses.

- Insightful Sentiment Analysis: Analyzing tweets provided valuable insights into public sentiment and protest trends, enriching the understanding of protest dynamics.

LIMITATIONS:

- Class Imbalance: The dataset's class imbalance, especially for Control Measures, impacted the model's prediction accuracy and led to higher misclassification rates.

- Data Quality and Features: Variability in data quality and limited features constrained the model's ability to fully capture the complexity of protest dynamics and state responses

- Lack of Real-Time Data: The model relies on historical data, which may not reflect current protest dynamics and state responses accurately.

DEPLOYMENT

These are our deployment sites:

- **Protest Outcome Prediction Model** https://protests.streamlit.app/

- **Mass Mobilization Analytics** https://mass-protests.streamlit.app/

thank you